

INPUT

Gene/protein or metabolite maker list



Or



Pathway enrichment analysis results

KEGG

MAPK signalling pathway (hsa04010)

positive regulation of **MAPK** cascade (GO:0043410)

regulation of **ERK1 and ERK2 cascade** (GO:0070372)

ERK1 and ERK2 cascade (GO:0070371)

Redundant pathways

Reactome

MAPK1/MAPK3 signaling (R-HSA-5684996)

MAPK family signaling cascades (R-HSA-5683057)

MAPA PROCESS

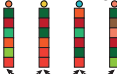
KEGG GO Reactome



Pathway name + description

Text embedding model (LLM)

Pathway embeddings



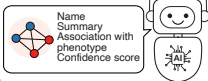
Pathway similarity matrix



Functional modules



Module annotation



OUTPUT

Summary



Functional module 1
Name: xxx
Summary: xxx
Association with: xxx
Confidence score: 0.95

Module network



Module heatmap

